power

File: power.kicad_sch

walter

File: walter.kicad_sch

usb_serial

File: usb_serial.kicad_sch

display

File: display.kicad_sch

peripherals

File: peripherals.kicad_sch

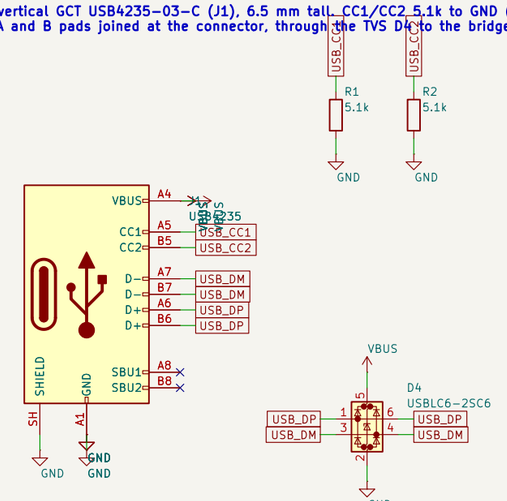
**awong-dev/pager**

Sheet: /
File: pager-carrier.kicad_sch

***Title: pager-carrier***

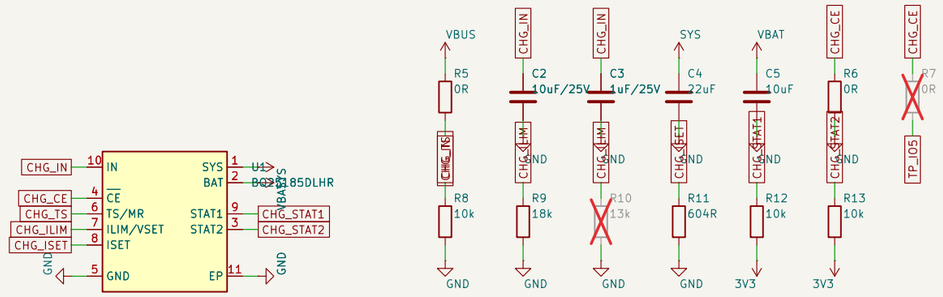
| Size: A4 | Date: 2026-10-17 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/6 |

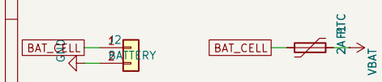
USB-C: vertical GCT USB4235-03-C (J1), 6.5 mm tall. CC1/CC2 5.1k to GND (UFP).
D+/D- A and B pads joined at the connector, through the TVS D4 to the bridge U5.



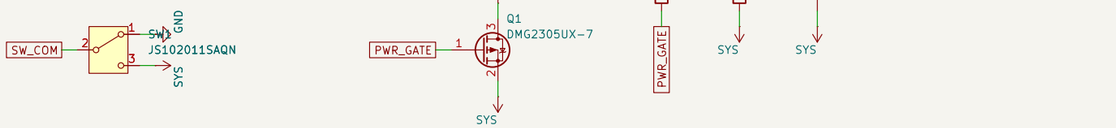

D4: flow-through TVS. Connector pads into pins 1/3, out of pins 6/4 toward U5.

Charger BQ25185DLHR: ISET 604R = 497 mA fast charge. ILIM/VSET 18k = 500 mA input limit, 4.20 V (13k DNP = 1.1 A).
TS unused: 10k to GND. CE is active-low: 0R to GND. STAT1/2 open-drain, 10k to 3V3.

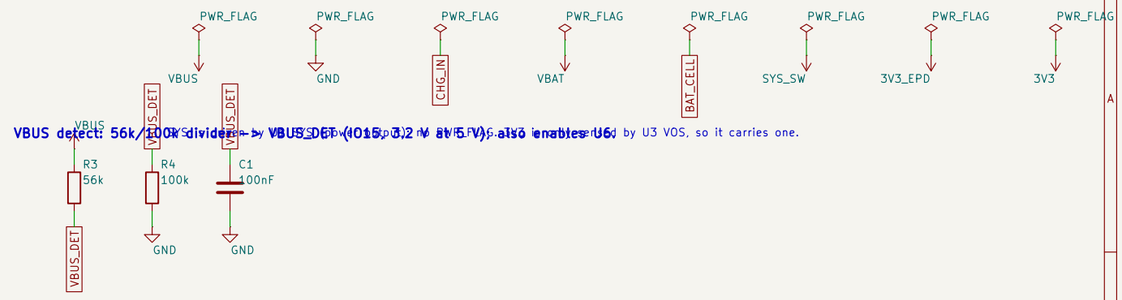



Battery: J2 (pin 1 = +) through F1 2 A PTC to VBAT. Switch: SW1 drives only the Q1 gate; SYS_SW = SYS when on.
2.2M gate pull-up = off when floating (1.9 uA through the ON switch, was 38 uA with 100k); 10k + 22nF soft-start limits inrush into the Walter bulk caps.

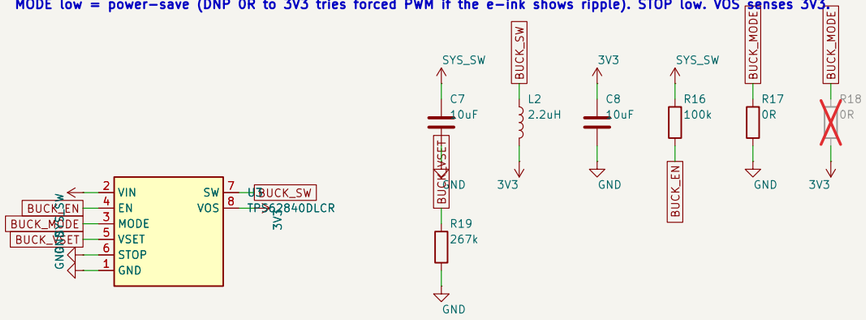



SW1: common (pin 2) -> SW_COM. Pin 1 -> GND = ON. Pin 3 -> SYS = OFF.

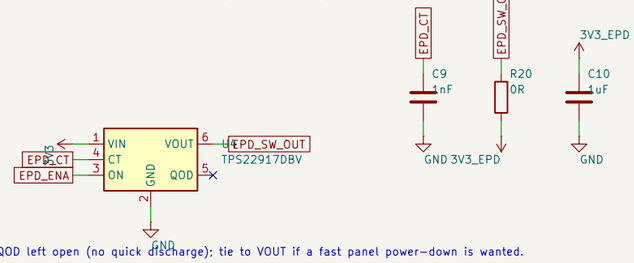



VBUS detect: 56k/100k divider -> VBUS_DET (1015, 3.2 V at 5 V), also enables U6.

3.3 V always-on rail: TPS62840DLCR buck from SYS_SW, 2.2uH, 10uF in/out. EN from SYS_SW via 100k. VSET 267k = 3.3 V.
MODE low = power-save (DNP 0R to 3V3 tries forced PWM if the e-ink shows ripple). STOP low. VOS senses 3V3.

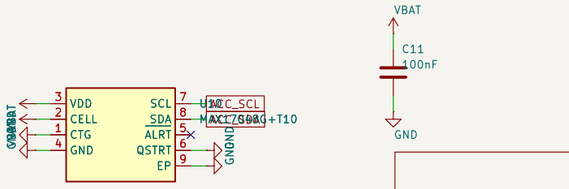



Display rail: U4 TPS22917 load switch 3V3 -> 3V3_EPD, ON = EPD_ENA (IO12), CT 1nF (~1 ms rise).
R-EPD 0R in series to measure panel standby current. QOD open.




QOD left open (no quick discharge); tie to VOUT if a fast panel power-down is wanted.

Fuel gauge U10 MAX17048 on VBAT (before the switch: keeps tracking while off, ~3 uA). I2C1 (ACC_*), addr 0x36. QSTRT and CTG low.

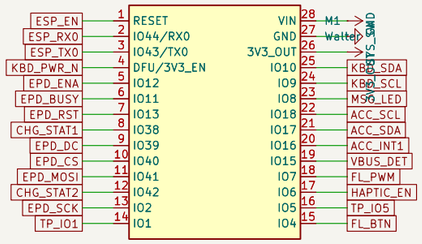


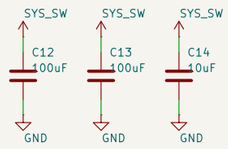

awong-dev/pager
Sheet: /power/
File: power.kicad_sch

**Title: USB-C input, BQ25185 charger, battery switch, 3.3 V rails**

| Size: A3 | Date: 2026-10-17 | Rev: A |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 2/6 |

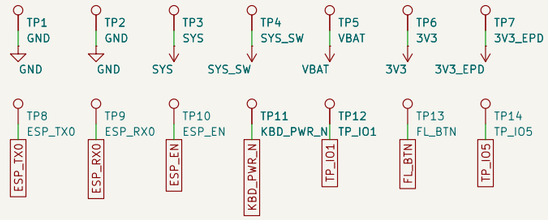
Walter 2x14 header. Pin names as datasheet rev 1.1 Table 1. The module is mounted header-side-up under the carrier; the footprint is the mirror of the vendor top-mount pattern.

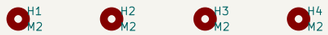



Bulk at Walter VIN (pin 28): 2 x 100uF 0805 + 10uF 0603, within 5 mm of the pin.

Test pads, 1.0 mm, top side.



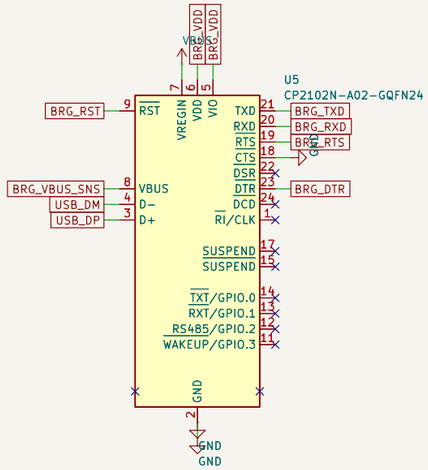



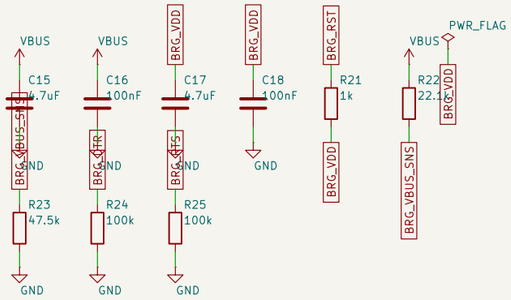

IO0 (pin 4, KBD_PWR_N) is a strap, the keyboard-rail gate and the auto-reset boot target; see usb_serial.

Spare IO1/IO5 go to test pads only (TP_IO1 = ADC1_CH0). IO5 also has a DNP OR to the charger CE. IO4 (RTC GPIO) = FL_BTN, the front-light button, see peripherals.

awong-dev/pager
Sheet: /walter/
File: walter.kicad_sch
Title: Walter module header, bulk capacitance, test pads, mounting holes
Size: A3 | Date: 2026-10-17 | Rev: A
KiCad E.D.A. 10.0.6 | Id: 3/6

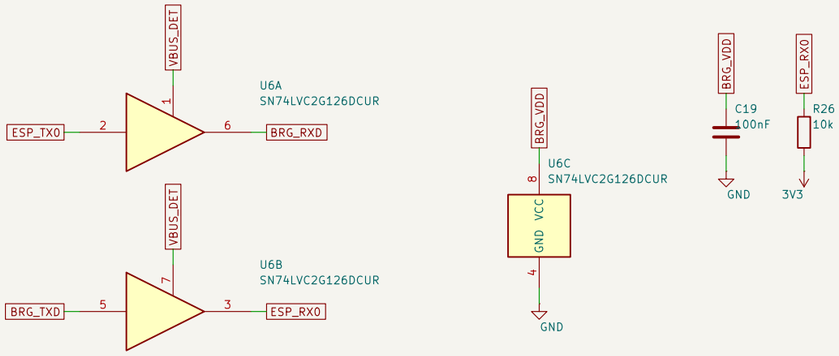
**USB-serial bridge: bus-powered CP2102N (unpowered without a cable), SN74LVC2G126 isolates UART0 when VBUS_DET is low, Espressif two-transistor auto-reset.**



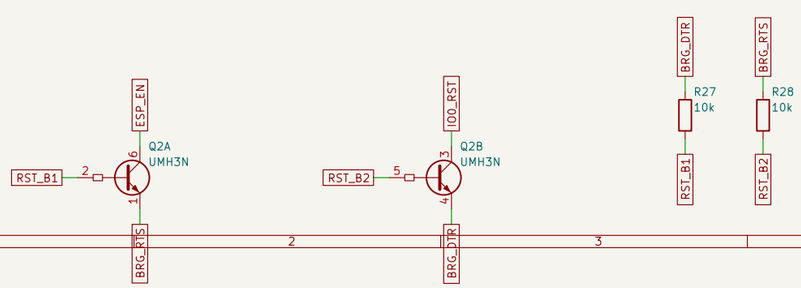




Bus-powered: VREGIN from VBUS, VBUS sense pin via 22.1k/47.5k, VDD = internal 3.3 V regulator output, VIO tied to VDD, 4.7uF + 100nF per power pin.
Program the USB product string 'Pager' with Simplicity Studio Xpress Configurator after assembly.

**U6 SN74LVC2G126: both OE from VBUS_DET. A: ESP_TX0 -> BRG_RXD. B: BRG_TXD -> ESP_RX0. 10k keeps ESP_RX0 high with no cable.**
**U6 VCC is BRG_VDD (bus-powered): with no cable the buffer is unpowered (Ioff outputs), so no LVC quiescent/float current on the battery.**

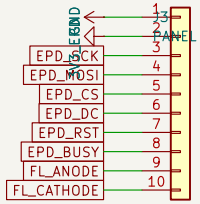


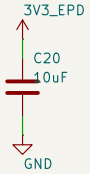

**Auto-reset (Espressif): Q2 DTR->EN, Q3 RTS->IO0. JP1 (closed) in Q3's collector lets auto-boot be disabled; reset still works.**




awong-dev/pager
Sheet: /usb_serial/
File: usb_serial.kicad_sch
**Title: CP2102N USB-UART bridge, isolation buffer, auto-reset**
Size: A3 | Date: 2026-10-17 | **Rev: A**
KiCad E.D.A. 10.0.6 | Id: 4/6

The SSD1680 ZIF, boost/charge-pump cluster and the front-light 6-pin ZIF live on the separate panel board (hardware/carrier/panel). The carrier carries only the 10-pin JST SH cable connector J3.




Pinout: 1 3V3_EPD, 2 GND, 3 EPD_SCK, 4 EPD_MOSI, 5 EPD_CS, 6 EPD_DC, 7 EPD_RST, 8 EPD_BUSY, 9 FL_ANODE, 10 FL_CATHODE. The panel board carries the RST/CS pull-ups, the VCI decoupling and the front-light ZIF.

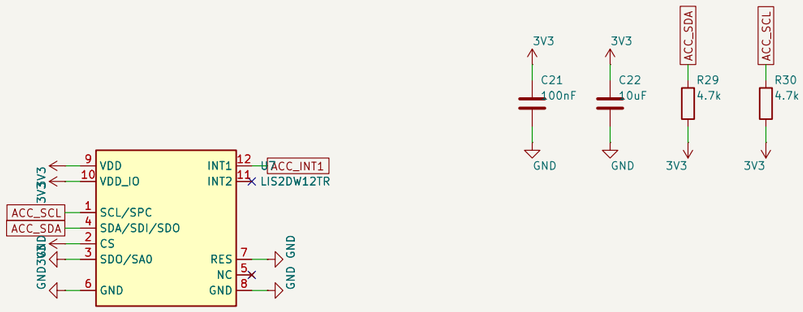
**awong-dev/pager**

Sheet: /display/
File: display.kicad_sch

***Title: Display cable to the panel board***

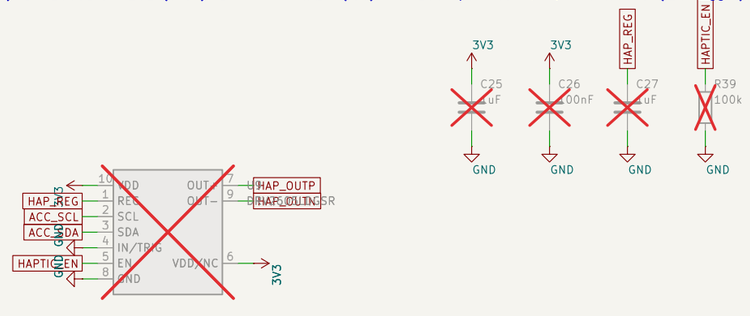
| Size: A4 | Date: 2026-10-17 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 5/6 |

U7 LIS2DW12 on I2C1 (ACC_*), SA0 low = 0x18, CS high = I2C mode, INT1 -> IO16 (RTC wake). Always on from 3V3.

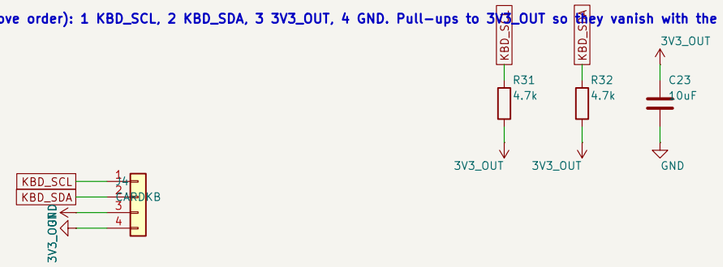



Place U7 away from J7 and the panel cable; X axis along the tin's long edge; mark orientation on silk.

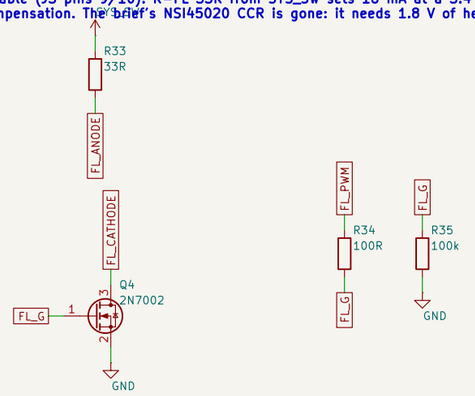
Haptic (DNP): DRV2605L on I2C1 (0x5A), EN = HAPTIC_EN (IO6) with 100k pull-down, IN/TRIG to GND (I2C trigger), OUT+/- to J7.



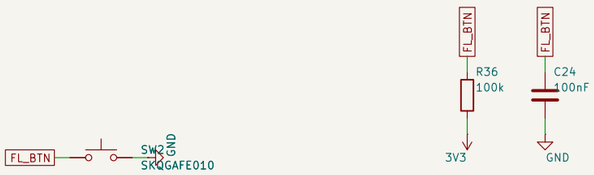

J4 CardKB (Grove order): 1 KBD_SCL, 2 KBD_SDA, 3 3V3_OUT, 4 GND. Pull-ups to 3V3_OUT so they vanish with the rail.




Front light GDEY029T94-FL03: 4 white LEDs in parallel on its own 6-pin 0.5 mm flex (1-2 anode, 3-4 NC, 5-6 cathode), Vf 2.75-3.3 V, 60 mA max.
The 6-pin ZIF sits on the panel board; anode/cathode ride the panel cable (J3 pins 9/10). R-FL 33R from SYS_SW sets 18 mA at a 3.4 V cell, 42 mA at 4.2 V, 50 mA on USB (SYS 4.45 V).
Q4 low-side switch, PWM from FL_PWM (IO7) for dimming / battery compensation. The brief's NSI45020 CCR is gone: it needs 1.8 V of headroom and was a 20 mA part.




SW2 front-light button: FL_BTN (IO4, RTC GPIO, ext1 wake-capable) to GND, 100k pull-up to the always-on 3V3, 100nF debounce. No current while idle.




awong-dev/pager

Sheet: /peripherals/
File: peripherals.kicad_sch

**Title: Accelerometer, keyboard, front light, message LED, haptic**

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-17 | **Rev: A** |
| KiCad E.D.A. 10.0.6 | | Id: 6/6 |